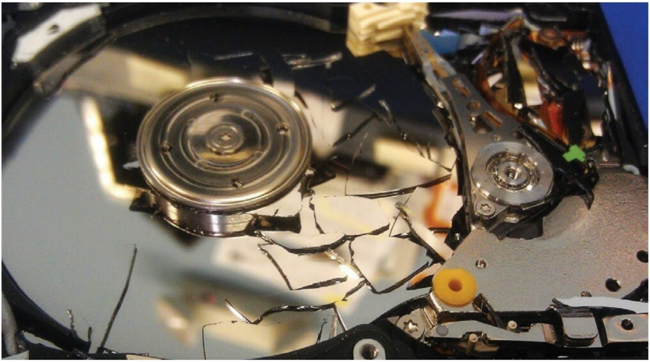
Automate the Bare Metal Provisioning Process through Razor

This article describes how to automate the bare metal provisioning of physical and virtual machines through Razor, an open source tool that works perfectly with Puppet.



Razor was created to automatically discover bare metal hardware and dynamically configure operating systems and hypervisors. Razor makes it easy to provision a node with no previously installed operating system and bring it under the management of Puppet. It was originally developed by EMC and is based on Tiny Core Linux. The Razor micro-kernel is 64-bit only. Razor can only provision 64-bit machines. It has the ability to discover hardware via in-memory instances of the Razor micro-kernel (a.k.a. Razor MK). The source code of the micro kernel is available at: *https://github.com/puppetlabs/razor-el-mk* under GPL v2 license.

Razor is completely open source, which means that it offers you the freedom to build your own custom Razor MK images, with the option to specify user accounts, the ability to enable remote SSH access for debugging, and to build and include custom Tiny Core Linux extensions to support unique hardware for your environment. Razor's policy-based bare-metal provisioning lets you make an inventory and manage the lifecycle of your physical machines.

How does Razor work?

Whenever a new node gets added, Razor discovers its characteristics by booting it with the Razor micro-kernel and inventorying its facts. The node is tagged based on its characteristics. Tags contain a match condition—a Boolean expression that has access to the node's facts, and determines whether the tag should be applied to the node or not. Node tags are compared to tags in the policy table. The first policy with tags that match the node's tags is applied to the node.

Provisioning elements of Razor

Repositories: These take care of 'What to install?' They basically indicate the contents to be installed on a system. To create a repository, either import or install an ISO or point at an existing package repository.

Tasks: A task takes care of 'How to install' using installation scripts such as kickstart files, preseed files and additional shell scripts. Predefined tasks are shipped with Razor, and custom tasks can easily be added without additional coding.

Broker: This takes care of 'How to manage' with post-installation scripts that install a configuration management agent on the node and enrol the node with the configuration management system (e.g., Puppet).

Tag: This takes care of 'Where to install' with the Boolean